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ARKbase: Antimicrobial Resistance Knowledgebase[1.0]

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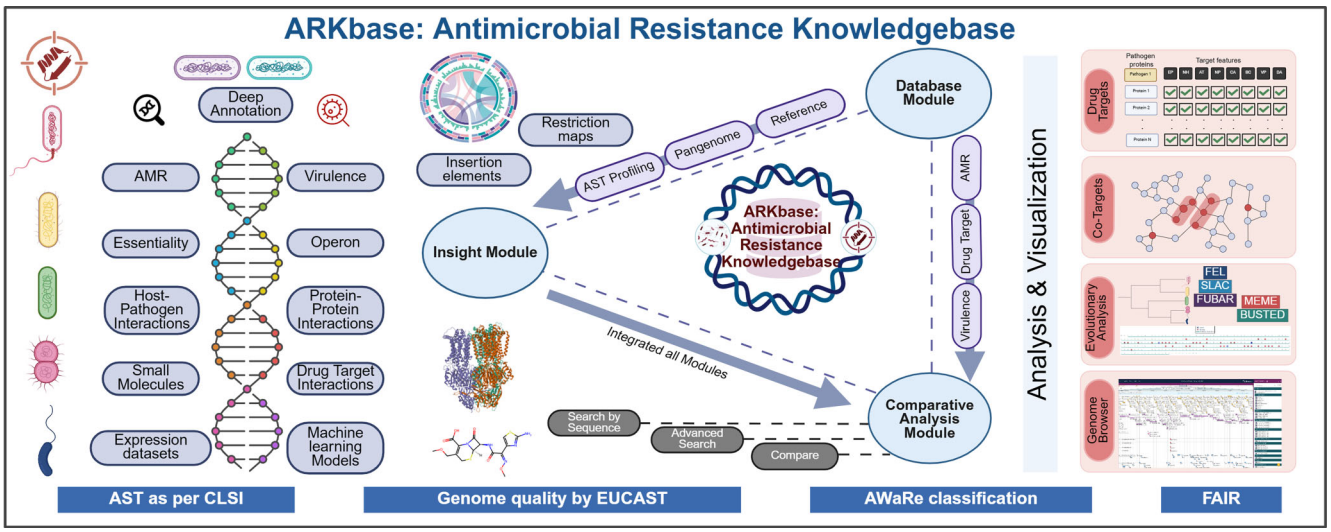
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Abstract

Antimicrobial resistance (AMR) is one of the most pressing global health challenges, often referred to as a 'silent pandemic.' Towards this, ARKbase is developed as an integrated, curated, value-added knowledgebase for AMR, with focus on WHO Bacterial Priority Pathogens. ARKbase has three core modules namely—Database, Insight, and Comparative Analysis Module—each offering curated data, including data analysed with custom-built pipelines. The Database Module has curated reference genomes and pan genomes along with antimicrobial susceptibility testing (AST) profiles. The Insight Module has 14 sub-modules and offers deep annotations encompassing antimicrobial resistance genes, virulence factors, structure annotation, host–pathogen interactions, biosynthetic gene clusters, known antibiotics, protein–protein interactions, drug targets, co-targets, drug-target interactions, machine learning models, and curated transcriptomic datasets for antibiotic exposure. The Comparative Analysis module offers a simple interface for comparing antimicrobial resistance genes, virulence factors, and drug targets among different priority pathogens. The data included in ARKbase are quality-checked and curated as per CLSI standards for AST, EUCAST guidelines for genome sequence, and the FAIR data principles. ARKbase is the first comprehensive knowledgebase focusing on WHO priority pathogens and AWaRe classification, enabling a combined evidence approach towards a holistic understanding of AMR. Available at https://datascience.imtech.res.in/anshu/arkbase.

Graphical abstract



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Introduction

Antimicrobial resistance (AMR), especially bacterial AMR, is one of the most pressing global health challenges. It is often referred to as a 'silent pandemic' as it generally goes unnoticed but has a devastating impact on human health [1]. Resistance is reported for all antibiotics currently in clinical use with only limited novel drugs in the pipeline. It is therefore imperative to understand the molecular mechanisms of AMR utilized by bacteria to explore novel strategies to combat resistance. The enormous impact and challenge of AMR is highlighted by the number of bacterial AMR-associated deaths, reported to be ∼5 million in 2019 and projected to rise to 10 million annually by 2050 due to a global increase in resistance rate [2, 3].

As early as 2017, the World Health Organization (WHO) released the first Bacterial Priority Pathogens List (BPPL) to guide discovery, research, and development of new antibiotics for drug-resistant bacterial infections. WHO has progressively refined their prioritisation scheme and more recently released the BPPL 2024 based on mortality, transmissibility, treatment options and several other criteria [4]. This list has 24 antibiotic resistance pathogens from 15 families, categorized into three priority levels—critical, high, and medium. The bacteria from these three groups present an array of intrinsic, adaptive, and acquired resistance mechanisms to reduce the antibiotic efficacy and evolve into multi-drug-resistant pathogens. The BPPL 2024 represents the most imminent threat to human health worldwide, highlighting the critical need for deeper understanding of AMR mechanisms that can be utilized towards development of novel therapeutics.

Over years, several databases have been developed that allow information retrieval and facilitate antibiotic discovery efforts. Many available resources offer in-depth information, such as CARD [5] and MEGARes [6] for antimicrobial resistance genes (ARGs), VFDB [7] for virulence factors (VFs), MIBiG [8], antiSMASH [9] for biosynthetic gene clusters (BGCs), etc. A few widely accessed resources are BV-BRC [10] and the Antibiotic Susceptibility Test (AST) Browser (https://www.ncbi.nlm.nih.gov/pathogens/docs/ast/). These resources offer large genome repositories with detailed functional annotations. Other databases are organism-specific and offer detailed insights in a species-specific manner [11]. However, to the best of our knowledge, there is no resource that is focused on WHO priority pathogens spanning all aspects of AMR research and antibiotic discovery pipelines providing a comprehensive integrative resource to direct drug discovery and research efforts for high-impact pathogens.

To this end, we present ARKbase (http://datascience.imtech.res.in/anshu/arkbase/), an integrated, curated, value-added knowledgebase and analysis platform, developed with the objective of providing a dedicated unified resource for Bacterial Priority Pathogens as identified by WHO. It is a unique BPP (Bacterial Priority Pathogen)-specific data resource and offers deep annotations for curated genomes encompassing ARGs, VFs, structure annotation (ST), host–pathogen interactions (HPIs), BGCs, protein–protein interactions (PPIs), drug-targets (DTs), co-targets, evolutionary constraint analysis, drug-target interactions (DTIs), curated machine learning (ML) models, and transcriptomic datasets for antibiotic exposure, etc. The strains included in ARKbase are quality checked and curated as per Clinical & Laboratory Standards Institute (CLSI) standards for antimicrobial sensitivity profiles (https://clsi.org/shop/standards/m100/) and European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines for genome sequence datasets [12]. As much as possible, data in ARKbase is classified and represented in the context of AWaRe classification [13] (Access, Watch, and Reserve) to facilitate understanding the AMR mechanisms in the backdrop of quantitative and qualitative assessment of the developing resistance against the antibiotic. As per this classification, Access antibiotics have low risk of resistance. Watch antibiotics are recommended for use in a limited number of infections and have higher potential for resistance, while the Reserve category antibiotics should be prescribed sparingly and only as a last resort, e.g. cases with multi-drug-resistant infections, and must be carefully monitored. ARKbase offers several unique features, including implementation of CLSI/EUCAST standards for reporting AST and genome datasets, identification of co-targets, providing a structured database for ARGs, offering a curated list of AI/ML models as per AWaRe classification, operon mapping of AMR and virulence genes, systems-level data for HPIs, and PPIs for WHO priority pathogens. ARKbase is a uniquely designed resource with carefully curated datasets to assist the community in exploring and innovating better strategies to combatAMR.

Data collection and curation

Fourteen organisms with species-level information from WHO BPPL 2024 were identified and included in ARKbase [14] (Supplementary Table S1). To curate genomes for each priority pathogen with antimicrobial susceptibility, the bulk genomes for these priority pathogens were retrieved from AST Browser (https://www.ncbi.nlm.nih.gov/pathogens/docs/ast/) and BV-BRC [10]. Three primary filters were applied for selecting data for integration in ARKbase, namely, host species as *Homo sapiens*, number of contigs ≤100, and the availability of AST profiles. The genome assemblies were then quality-checked using QUAST [15] as per the EUCAST WGS guidelines—contig count ≤100, NG50 >30 000 bp, and ≤100 contigs with sequence lengths >500 bp [12]. For AST, phenotypic profiles of the quality-checked genomes were evaluated as per the CLSI breakpoint criteria. AST data are presented as per the AWaRe classification criteria [13] (details of all steps involved for curation of genomes and phenotype data are provided in Supplementary data). These quality-assessed genomes (>10 000) (Supplementary Table S2) were annotated using PGAP [16] and these annotations are consistently used across ARKbase to report annotations at the gene level. Further curation of data for each module is discussed in their respective sections (Supplementary Table S3).

ARKbase modules

All available information in ARKbase is represented in three seamlessly integrated core modules—Database module, Insight module, and the Comparative Analysis module. These modules are accessible through the home tab on ARKbase. Each module has several sub-modules allowing user-friendly navigation, database querying and retrieval and analysis of data. To ensure ease of use, the modules are supported by graphical abstracts. A schematic representation of ARKbase core modules, submodules, workflow data sources, and functionality is illustrated in Fig. 1 and Table 1. We now discuss the core modules architecture and their utility.

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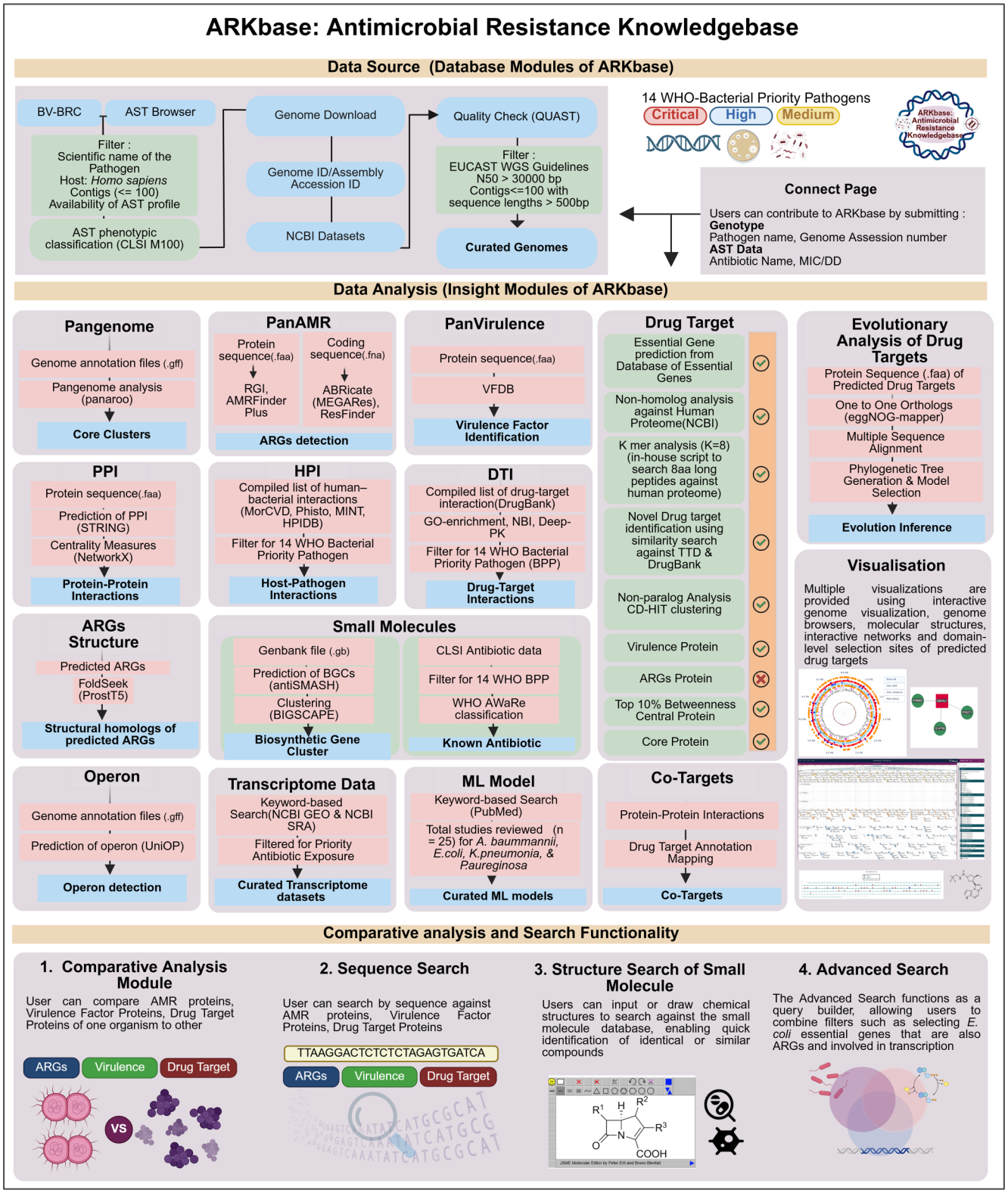


Figure 1. Schematic representation of the ARKbase workflow, illustrating each module with brief descriptions of the steps, along with the associated data sources and applications.

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Table 1. Databases and tools utilized in ARKbase modules

S. No.	Module	Database used	Tools used
1	Curated genome AST	BVBRC [10], AST Browser (https://www.ncbi.nlm.nih.gov/pathogens/docs/ast/), AWaRe classification [13]	QUAST [15], genome_fetch.py*
2	Operons	NCBI RefSeq [17]	UniOP [18]
3	COG(Cluster of orthologous genes)	eggNOG (v5.0) [19]	eggNOG-mapper (v2) [20]
4	Restriction mapping	REBASE (Restriction Enzyme Database) [21]	EMBOSS Restrict (v6.6.0) [22],restrict.pl*, restrict_to_gff3.pl*
5	Pan AMR	CARD (v 4.0.1) [5], ResFinder (2024) [23], AMRFinderPlus (16 July 2025) [24], MEGARes (v3) [6]	RGI (v4.0.1) [5], ResFinder(v4.7.2) [23], AMRFinderPlus (v4.0.23) [24], ABRicate (v1.0.1) (https://github.com/tseemann/abricate)
6	Pan virulence	VFDB (v2.0 set B proteins) [7]	BLASTp (v2.14.1) [25]
7	ARG (antimicrobial resistance gene) structure	PDB [26], AlphaFold (AF2) [27]	Foldseek (9.427df8a) [28]
8	PPI (protein–protein interaction)	NCBI RefSeq [17], STRING (v12) [29]	NetworkX (v3.5) [30]
9	HPI (host–pathogen interaction)	HPIDB 3.0 [31], MorCVD (4 October 2024) [32], MINT (21 October 2024) [33], Phisto (21 October 2024) [34]	1_Filtering_database.R*
10	Drug targets	VFDB (v2.0 set A proteins) [7], STRING (v12) [29], Database of Essential Genes (v15) [35], Therapeutic Target Database (v10.1.01) [36], DrugBank(v5.1.13) [37]	BLASTp (v2.12.0) [25], CD-HIT(v4.8.1) [38], MMseqs (v16.747c6) [39], kmer.pl*, extract_core.py*, Panaroo (v1.5.2) [40]
11	Evolutionary analysis	NCBI RefSeq [17], eggNOG [19]	NCBI E- utilities [41], eggNOG-mapper [20], ClustalW [42], PAL2NAL [43], IQTREE [44], FEL [45], FUBAR [46], SLAC [45], MEME [47], BUSTED [48], HyPhy [49], iTOL(v7) [50], EvolutionaryAnalysis*
12	Co-target	NCBI RefSeq [17], STRING (v12) [29]	NetworkX (v3.5) [30]
13	Small Molecules – Known Antibiotics	CLSI (https://clsi.org/), AWaRe classification (https://www.who.int/publications/i/item/WHO-MHP-HPS-EML-2023.04)	curate_antibiotics.R*
14	Small Molecules – BGC (Biosynthetic Gene Cluster)	NCBI RefSeq [17]	antiSMASH 8.0 [9], BIG-SCAPE 2.0 [51]
15	Small Molecules – Structure Search	PubChem [52]	JSME tool [53], structure_search.py*
16	DTI (drug-target interaction)	DrugBank (v 5.1.12) [37], QuickGO [54], CARD (v 4.0.0) [5], DEG (1 September 2020) [35]	nbi_main.ipynb*, BLASTp (v2.12.0) [25], Deep-PK [55]
17	Comparative analysis	ARGs, VFs, and DTs	BLASTp (v2.12.0) [25], pyCirclize (https://github.com/moshi4/pyCirclize), Plotly (https://github.com/plotly/plotly.py)
18	Genome Browser	NCBI RefSeq [17]	JBrowse 2 [56]

This table summarizes the databases and bioinformatics tools employed in the modules of ARKbase. Tools marked with an asterisk (*) indicate in-house or customized scripts available in their respective modules in ARKbase GitHub (https://github.com/AB-DataScienceLab/ARKbase/tree/main/ARKbase%20Analysis).

Database module

The Database module has two sub-modules, namely, 'Reference Genomes' and 'Curated Genomes AST.' Fourteen organisms with species-level information from WHO BPPL 2024 were identified and included in ARKbase [14]. These can be accessed from the Reference Genome sub-module, organized as critical, high, and medium priority pathogens. Selection of a specific pathogen loads the pathogen overview, genome summary, and representation for Clusters of Orthologous Genes (COG) classification, operons, ARGs, structural annotation of ARGs, VFs, PPIs, and DTs (Supplementary Fig. S1). An overall representation of antimicrobial susceptibility profiles of all genomes of this pathogen in ARKbase is provided and also mapped with AWaRe classification (https://www.who.int/publications/i/item/WHO-MHP-HPS-EML-2023.04). A visual representation in the form of selectable genome tracks can also be accessed through the genome browser tab.

For the 'Curated Genomes AST' sub-module, each isolate is grouped and displayed as per AWaRe classification and CLSI guidelines. In addition, antibiotics and antibiotic class, along with year of collection and country of isolation, are also listed (Additional Supplementary data). The data is linked to the source and can be queried and downloaded for further analysis.

Insight module

ARKbase's insight module is a composite module with 14 sub-modules. For ease of understanding, these sub-modules are functionally categorized and discussed in two sections—Basic annotation and Deep annotation. Basic annotation includes gene function, operon mapping, COG classification, Insertion Sequence (IS) element, and restriction site annotation. Deep annotations include ARGs, VFs, ARG structures, PPIs, HPIs, and pangenome analysis. The basic and deep annotations are done using several tools and are discussed in detail in the Supplementary data. The insight module's data summary page provides a summary of the deep annotations applied on

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the reference and pangenome data, including AMR, virulence, and COG categories.

Operons

Operons are spatially and functionally clustered sets of genes transcribed together under a single promoter typically encoding proteins with related biological functions in a specific pathway. AMR often results from co-ordinated expression of multiple genes rather than a single gene/mutation—including efflux pumps, regulators, transporters, and enzymes. For example, the mar (multiple antibiotic resistance) operon in *Escherichia coli* encodes transcriptional regulators like *marA* and *marR*, which control the expression of efflux pumps and porin channels. In addition, *marA* also activates genes involved in DNA repair and lipid trafficking, thereby enhancing survival under antibiotic pressure and reducing drug uptake [57]. Characterizing these gene clusters is fundamental to uncover the regulatory strategies bacteria use to respond to antibiotics and stress conditions. Identification and mapping of operons in WHO Priority Pathogens provide insights into co-regulated gene networks that contribute to AMR, virulence, and pathogenicity. Operon annotations can reveal previously uncharacterized genes linked to resistance phenotypes. Additionally, operons regulating both AMR genes and VFs can be potential therapeutic targets, as targeting a single regulating mechanism may affect a range of bacterial survival strategies. Integrating operon-level data into ARKbase offers a powerful approach to uncover hidden regulatory co-relations and drive forward the development of novel antimicrobial strategies. In ARKbase, operon prediction was performed using a customized UniOP pipeline [18] (Supplementary Fig. S3).

COG classification, IS elements, and restriction mapping

Functional annotations like COG classification can be used to understand AMR genes by grouping them based on their shared evolutionary ancestry and functional roles. In ARKbase, COG classification is performed using eggNOG-mapper (v2) [20] (Supplementary Fig. S4). Similarly, IS elements contribute to genome plasticity by virtue of their intrinsic ability to hop in and between genomes adding to genome dynamism via rearrangements. Their insertion *in cis* or within genes can result in loss or alteration of gene function, origin of novel genes and novel functions. They are also known to play a key role in development and spread of AMR via horizontal gene transfer. The IS elements were annotated using ISEScan with default parameters [58]. Restriction mapping tracks were generated by the Restrict utility in EMBOSS for restriction enzymes with unambiguous recognition sequences of length five or above [22].

AMR genes & VFs

ARGs form the functional core of AMR. The annotation of ARGs in microbial genomes is extremely critical, as these reflect the existing and evolving AMR strategies in the pathogen, which may help fine-tune our combat strategies against AMR. Given the criticality and to ensure that none of the putative ARGs are missed, we use four different resources namely, RGI (v4.0.1) [5], ResFinder (v4.7.2) [23], AMRFinderPlus (v4.0.23) [24], and ABRicate (v1.0.1) (https://github.com/tseemann/abricate) with MEGARes (v3.0) [6] for annotation of ARGs in ARKbase (Supplementary Table S4 and Supplementary Fig. S5). These annotations are available through the reference genomes tab for reference genomes and via the Pan AMR tab for curated genomes. In the Pan AMR tab, functionality is also provided to filter data based on specific antibiotics as well as phenotypes (resistant, susceptible, and intermediate). The ARG annotations are also made available for visualization through the genome browser.

VFs are specialized molecules that give pathogens the ability to cause infection and enhance their ability to cause diseases. VFs facilitate pathogens attaching to or entering host cells, inhibit or evade the host's immune response, and obtain nutrients through ecosystem engineering. They influence the persistence and severity of infections and therefore are critical in the context of AMR. ARGs and VFs often co-exist and share pathways—for example, the *pmrAB* two-component system mediates colistin resistance and immune evasion in *Acinetobacter baumannii* [59]. Their proximity in genomes and presence on mobile genetic elements promote co-transfer via horizontal gene transfer. Hence, studying ARG and VF is vital to curb resistance spread. Along with their biological significance, VFs are also important as potential DTs for novel therapeutics in the efforts to tackle AMR. In ARKbase, VFs are mapped using VFDB v2.0 [7] and can be visualized in the genome browser (Supplementary Fig. S6 and Supplementary Table S4).

ARG structures

Understanding protein structure is crucial for developing new drugs to combat AMR. Therefore, the structural annotations of ARGs in reference genomes are provided. The details include availability of known crystal structures; when not available, the structures are retrieved from AlphaFold [27]. In cases where structures are not available from RCSB or AlphaFold, FoldSeek [28] and ProsT5 [60] are applied to gain structural information (Supplementary Fig. S7).

Networks & interactions

A system is more than the sum of its individual parts. It is the dynamic interplay between the individual parts that provides the system its emergent properties. Hence, a deep annotation involving the networks and interactions in BPPs is absolutely imperative to our understanding of AMR and development of combat strategies [61–63]. In ARKbase, two insights modules describe network- and interaction-based data, namely, PPIs and HPIs.

Protein–protein functional interactions

The PPI networks of WHO BPPL provide an overall picture of how proteins interact and function together within the system. These interactions are represented as PPI networks. These networks can be analysed using established network analysis metrics like centrality values. These values are numerical measures of the importance or influence of a node within a network. High degree centrality indicates hub proteins, which interact with a large number of proteins and may play a central role in cellular functions. High betweenness centrality highlights proteins that mediate inter-pathway communication, acting as critical connectors or bottlenecks. Meanwhile, high closeness centrality reflects proteins that have greater accessibility to all other nodes, enabling faster information flow across the network. According to the centrality-lethality hypothesis, high central proteins (especially betweenness central-

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ity) are very critical for survival, and targeting them disrupts the whole system [64]. These proteins are likely to be viable DTs. For ARKbase, reference proteomes of all 14 pathogens were retrieved from NCBI datasets and PPI data were retrieved from STRING v12 [29]. The interactions were filtered for high-quality interactions (combined scores ≥700). To identify the critical proteins, different centrality measures—betweenness, degree, and closeness—have been calculated using NetworkX [30]. Subsequently, these centrality values were classified into four groups—Q1 (top 25%), followed by Q2, Q3, and Q4 (Supplementary Fig. S8). The PPI tab in ARKbase provides a consolidated list of these interactions with centrality values and also maps the interactor proteins to their COG category [65]. The tab also provides functionality to select organisms of interest, proteins of interest, and also the range of centrality values. A separate tab is provided to explore co-targets based on this data and is discussed in the Drug and Targets module below.

Host–pathogen interactions

HPIs play a vital role in understanding the role of how pathogenic bacteria interact with human cellular machinery during infection. HPI data for the 14 pathogens was curated from four established interaction databases—HPIDB [31], MorCVD [32], MINT [33], and Phisto [34], filtered for interactions between human and BPP proteins (Supplementary Fig. S9). This module enables identification of proteins that are at the intersection of being essential for pathogen survival and critical for host interaction. A protein that is both essential and directly engages with the host represents a potential DT, as it is involved in both pathogen survival and also in the mechanism of infection in the host. By linking functional classifications (COG) [65] for the targeted human proteins, the module helps in determining which host cellular processes are commonly exploited by different pathogens for infection and their survival. The HPI tab also presents the top 10 GO term distributions of pathogens and an interactive explorer with searchable tables and network visualization for each BPP.

Pangenome analysis

Pangenome construction offers a comprehensive view of genetic variation at both strain and species levels. Pangenome analysis is essential for understanding the dynamic and emerging genetic landscape of a species, capturing both the conserved core genes that underpin fundamental biology and the accessory genes that drive diversity, adaptation, and AMR. The pangenome data along with AST profiles facilitate genotype-phenotype correlations in a species-specific manner. There are two modules, as discussed earlier, that provide pangenome-based annotations: Pan AMR and Pan Virulence. The Pan AMR module gives a combined view of identified ARGs in a tabular format along with their description and the source of their annotation. The users can download this table in .csv format for each priority pathogen. The 'Pan Virulence' module offers descriptive information on the identified VFs in an organism-specific way that can also be downloaded for further analysis.

Panaroo [40] was employed for pangenome construction and core genes for each priority pathogen were identified from the gene presence–absence file, considering those present in 99%–100% of total strains (Supplementary Fig. S2). A representative sequence from each core gene cluster was selected and used for subsequent analysis for identification of DTs.

Drug & target module

In this section, four Insight modules are included—DTs, Co-targets, Small Molecules, and DTIs. The DTs module is further subdivided into two sections—(a) identification of DTs and (b) evolutionary analysis of prioritized DTs identified. The small molecules module is further subdivided into three submodules: known antibiotics as reported in CLSI, BGCs, and small molecule structure search utility against antibiotics and secondary metabolites derived from BGCs.

Drug targets

ARKbase offers a unique and comprehensive functionality for prioritisation of potential DTs. Each protein in a given proteome is classified in terms of essentiality (defined by sequence homology with DEG [35] - MMSeqs2 [39] with e-value $<1e^{-5}$), human non-homolog (no similarity with human proteome - MMSeqs2 with e-value $<1e^{-4}$), Kmer (eight amino acid long kmer with no exact peptide matches with the human proteome - in-house Perl script), non-paralogs (identified by CD-HIT clustering [38] with identity ≥0.8), VF (MMSeqs2 with e-value threshold of $1e^{-5}$ with VFDB [7]), central protein (based on betweenness centrality values), core (pangenome core), and ARG (Supplementary Fig. S10). By default, the data is presented for the entire proteome of each species against different features mentioned earlier. In addition, the novelty of potential DTs can be evaluated based on the absence of this protein in the therapeutic targets database (TTD), presented as TTD Novel [36], and in DrugBank [37], presented as DrugBank Novel. The user can choose to opt for various combinations of these criteria to identify proteins of interest. We have defined the prioritized DTs as the proteins that fulfilled all the criteria, following a filtering approach.

Evolutionary analysis of selected DTs

For DTs selected earlier, evolutionary analysis is performed in ARKbase, and the data are provided in the DTs module. Target proteins under purifying selection indicate that these proteins may have limited scope of adaptation and therefore may serve as priority candidates for drug discovery. The details of each step are discussed in the Supplementary data (Supplementary Fig. S11). In the DTs module, on selecting an organism of interest, the user can access the evolutionary analysis tab that provides details on the protein-level and site-level constraints. The DTs are annotated for domains with InterPro and domains are underlined in the accompanying visualization [66]. The sites under positive and purifying selection as predicted by different methods are highlighted in separate colours and shapes for ease of data interpretation.

Co-targets

The co-target sub-module can be accessed directly through the co-targets tab and also through PPIs tab. It offers exploration for identification of potential co-targets. Each interactor in this sub-module is provided with its protein description, betweenness centrality value, and COG description. It is proposed that top central proteins, which are also interacting partners, offer an alternative approach for co-target identification (Supplementary Fig. S8). This is a unique feature of

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ARKbase as centrality-based assessment and identification of co-targets is not a routine approach. ARKbase offers this feature in an easy user-friendly interface where users can also navigate drug-like properties of potential co-targets and enable prioritization for polypharmacology.

Small molecules

This sub-module is further divided into three sections: known antibiotics as reported in CLSI, BGCs, and small molecule structure search.

Known antibiotics

The known antibiotics tab provides a curated library of antibiotics with their minimum inhibitory concentration (MIC) breakpoint values for the BPPs sourced from the CLSI (Supplementary Fig. S12a). This data is integrated with the WHO AWaRe classification (Access, Watch, Reserve) to map the antibiotics with their respective AWaRe class (https://www.who.int/publications/i/item/WHO-MHP-HPS-EML-2023.04). In addition, for each antibiotic, the CLSI breakpoints, structure, and SMILES are also provided. The known antibiotics tab has utility to filter antibiotics by pathogens and within each pathogen by AWaRe classification. The entire data is available for download for further analysis.

Biosynthetic gene clusters

BGCs in a genome refer to the presence of two or more genes in a group that code for a biosynthetic pathway resulting in the production of specialized or secondary metabolites. These metabolites of microbial origin are known to have medicinal properties and applications in agriculture. Numerous dedicated resources like antiSMASH [9], MiBIG [8], and BiG-SCAPE [51] are available for annotation of BGCs. In this study, these resources have been used to analyse the BGCs of BPPs to find (a) chemical-scaffold-producing BGCs with antimicrobial potential and (b) any BGC which produces a metabolite crucial for pathogenesis, virulence, or biofilm formation in BPPs that may be explored as potential DT (Supplementary Fig. S12b). The BGC secondary metabolite library is also provided for download.

Structure search

The Structure Similarity Search tool is a feature designed to compare the structure of a user-provided query compound against two curated chemical libraries—(i) Known Antibiotics and (ii) Predicted BGC Metabolites. In cases where the queried structure has similarity to a known antibiotic, it may suggest that the queried compound shares a similar mechanism of action. This may also indicate that the queried compound is a known scaffold and other scaffolds can be probed for novelty. In case the user's queried structure is similar to a BGC metabolite, it may indicate that the pathogen already produces a similar compound. This might imply an intrinsic resistance mechanism, suggesting the queried compound may be less effective against that particular pathogen, and these structures may be deprioritized in antimicrobial drug discovery initiatives.

Drug-target interactions

The fourth sub-module is Drug-target interaction (DTI), which provides novel, computationally predicted drug repurposing candidates against WHO's priority pathogens. In this analysis, existing DrugBank [37] interactions are utilized to predict new DTIs. For this, the interactions are enriched using Gene Ontology and analysed using a Network-Based Inference (NBI) approach [67, 68]. The predicted DTIs are subsequently subjected to a stringent filtering cascade for prioritisation of potential drug and target candidates (Supplementary Fig. S13). The DTI tab presents the top 10 GO term distributions of pathogens and an interactive explorer with searchable tables and network visualization for each BPP. This module offers novel scaffolds and alternatives to existing antibiotics bypassing the challenges of traditional drug discovery approaches.

Curated ML models & expression datasets

The ML module and expression (transcriptomics) module have curated data for the BPPs. The ML module offers the user reported ML models for different antibiotics divided into AWaRe classes, along with the link of their GitHub repositories, wherever available. ML is rapidly emerging as a powerful approach for predicting AMR and MICs from genomic and transcriptomic data. However, published ML studies vary greatly in pathogens studied, algorithms applied, features used, performance metrics, and the availability of code or datasets, making it challenging for researchers to identify relevant work, compare methods, and reuse existing resources. The ML Models Curation module in ARKbase addresses this gap by systematically summarizing key aspects of published ML-based AMR prediction studies. It captures details on pathogens covered, algorithms used (e.g. random forest, XGBoost, deep learning), model performance (accuracy, F1-score, RMSE, or other reported metrics), and data availability (GitHub repository link or article link if code or data are unavailable). Models are organized by targeted antibiotics and classified according to the WHO AWaRe categories, enabling users to quickly locate studies relevant to specific pathogen–drug combinations (https://www.who.int/publications/i/item/WHO-MHP-HPS-EML-2023.04). By providing structured, comparable, and accessible information, this module streamlines literature review, facilitates method selection, supports reproducibility, and serves as a valuable starting point for any AMR research—whether developing new predictive models, validating existing ones, identifying knowledge gaps, or integrating computational tools into clinical and surveillance workflows (Supplementary Fig. S14).

Transcriptomic studies not only reveal the dynamic adaptation of pathogens to antimicrobial pressure but also differentiate between the mere presence of resistance genes and their active expression, providing a more functional perspective on resistance. Using expression data, researchers can identify resistance-associated genes, including efflux pumps, antibiotic-modifying enzymes, and target modification pathways, as well as elucidate global regulatory networks and stress response mechanisms activated during antibiotic exposure. Furthermore, integrating transcriptomic data with other omics approaches facilitates the discovery of novel therapeutic targets, improves our understanding of HPIs, and supports the development of effective surveillance strategies for AMR. In this module, we provided information about all the transcriptomics datasets available in the public repositories—NCBI Sequence Read Archive (SRA) [69] and Gene Expression Om-

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nibus (GEO) [70] for the BPPs with respect to antibiotic exposure (Supplementary Fig. S15).

Visualization & search modules

ARKbase offers several different visualizations ranging from CIRCOS plots for genome representation to PPI networks. It was felt necessary to provide a consolidated visualization of all possible annotations reported in ARKbase. The genome browser module provides an interactive visualization platform for exploring annotated genomic features associated with AMR (Supplementary Fig. S16). Built on the JBrowse 2 framework, this module enables users to navigate bacterial genomes of the priority pathogens with ease, facilitating detailed inspection of resistance genes, VFs, operons and other relevant genomic landmarks along with gene annotation [56, 71]. ARKbase offers two search utilities—'Search by Sequence' and 'Advanced Search.' Search by Sequence allows users to search a query by entering or uploading a protein sequence, which is then searched against one of the chosen ARKbase's databases—AMR, virulence, or DT proteins—through BLASTp [25] with the user-defined parameters [Supplementary Fig. S18(a)]. Each query generates downloadable BLASTp results along with the protein features of the subject protein for added information [Supplementary Fig. S18(b)]. Advanced Search or Query Builder allows the user to build complex queries by selecting pathogens and protein features with AND or OR operators [Supplementary Fig. S19(a)]. For example, users can identify proteins that are classified as essential, VFs, or ARGs for *E. coli* [Supplementary Fig. S19 (b)].

Comparative analysis module

The comparative analysis module in ARKbase is developed with the objective of providing some basic utilities that allow comparison of antimicrobial resistant genes, VFs, and DTs across BPPs. This module can be accessed through the main home tab and the search tab. The comparative analysis module page is divided into three sections. The first section offers a simple utility to select ARGs, VFs, or DTs in any species and perform sequence-based comparisons with other species. In the second section, precomputed all-to-all comparison results are provided for ARGs, VFs and DTs. In the last section, a CIRCOS-based visualization is provided for visual comparison between any two selected species for ARGs and VFs [71]. This interactive genome visualization tool allows users to draw comparisons between two different genomes in a visually interactive representation and hover the mouse over genes to show the annotated gene name, coordinate information, and product information. All datasets are readily available for download and can be utilzed for further downstream analysis (Supplementary Fig. S17).

Accessory modules & ARKbase architecture

In addition to the core modules, three additional tabs are provided—Connect (Supplementary Fig. S20), Data Summary (Supplementary Fig. S21), and FAQs & Helpfile.

The backend was built using the LAMP stack, which includes Linux (operating system), Apache (web server), MySQL (database), and PHP (server-side language). The frontend is created with HTML, CSS, and JavaScript. The ARKbase platform is supported by all modern web browsers.

Discussion: use cases and future plans

Use cases

To illustrate the utility of ARKbase, we now discuss three use cases (Fig. 2).

Use case 1: Prioritize potential DT candidates in priority pathogens that satisfy multiple desirable criteria (essential for pathogen survival, not homologous to human proteins, VFs, etc.).

To demonstrate this use case, we used the Advanced Search utility in ARKbase. We initiated a search by selecting *E. coli* as the pathogen and then applied filters using the protein feature option with AND operator. The protein feature filters included essentiality, non-human homolog, kmer, core protein, non-paralog, VF, top 10 central proteins, and novelty as per DrugBank and TTD. These filters yielded a total of 13 proteins, which can be considered as potential DT candidates.

Use case 2: Identify common DTs across priority pathogens for discovery of potential broad-spectrum inhibitors

We selected one of the top candidates in *Escherichia coli* from use case 1, UDP-*N*-acetylglucosamine acyltransferase (LpxA), and utilized the Search by sequence utility in ARKbase to look for homologs in other priority pathogens. As can be seen in Fig. 2, hits are obtained in several cases, with very high sequence similarity in at least four priority pathogens. In addition to sequence-level information, ARKbase also provides information on homologous proteins in respective pathogens for key parameters like essentiality, centrality, non-homology to human proteome, non-paralogy, role in virulence, and AMR. These properties in addition to sequence similarity offer a combined evidence approach towards identification of broad-spectrum DTs. Further, literature search also reports the significance of LpxA in lipid A biosynthesis pathway, which is considered an attractive DT against multi-drug-resistant Gram-negative bacteria, thus lending credence to the use case [72, 73].

Use case 3: Testing novelty of potential inhibitors

Extending the already discussed use case 2, novelty of potential inhibitors for LpxA was probed. For this, we utilized the structure search module of ARKbase. Two compounds identified as potential inhibitors for LpxA were used as a query in this module [74]. No hits were obtained with known antibiotics indicating potential new scaffolds.

These use cases indicate the integrative and combinatorial utility of ARKbase in framing and testing hypotheses seamlessly.

Future plans

ARKbase is a unique resource developed in response to the WHO Bacterial Priority Pathogen list (2024). A list of functionalities and data unique to ARKbase is presented in Supplementary Table S5. ARKbase has a modular design that can cater for inclusion of new modules and data updates.

As annotation is a continuous and evolving process, ARKbase offers utility for community curation. This is available via the Connect tab. The submission form accepts contributions for genotype and phenotype data related to AMR in WHO BPPL. For genotype data, contributors are required to

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Use case 1:

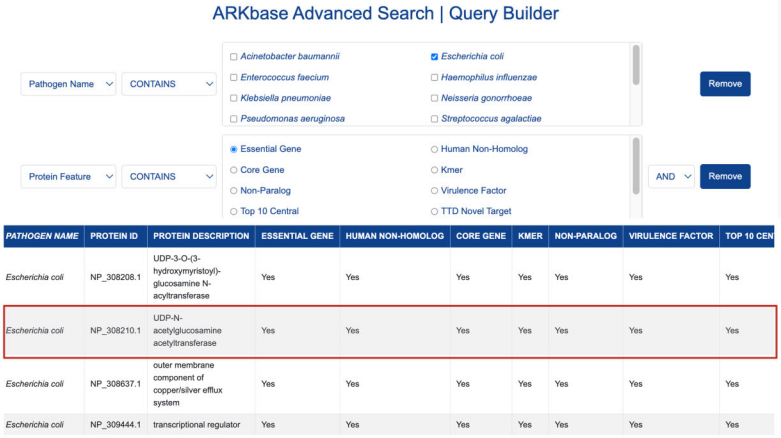
Pathogen name, e.g., *Escherichia coli*

↓

Apply Filters (AND operator)
- Essential gene
- Non-human homolog
- Core protein
- Non-paralog
- Virulence factor
- Top 10 central protein
- Novelty check (DrugBank + TTD)

↓

13 prioritized proteins (Potential drug target candidates)



Use Case 2:

Select one of the top candidate from Use Case 1 (e.g., UDP-N-acetylglucosamine acyltransferase, LpxA in *Escherichia coli)*

↓

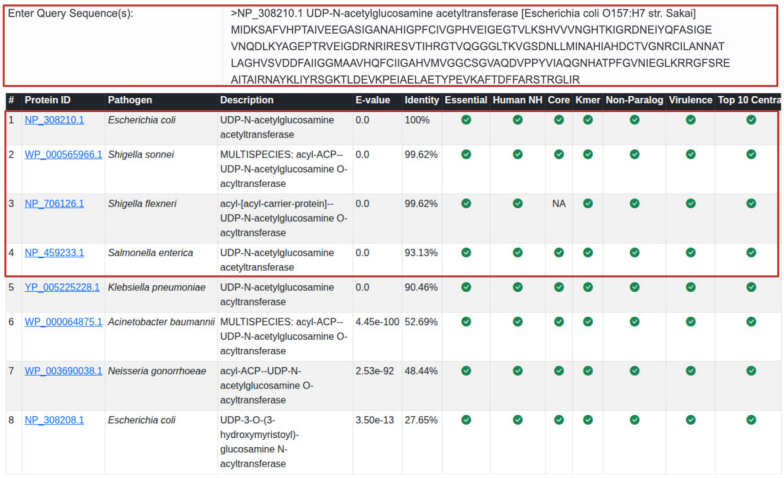
Cross-Pathogen Search (ARKbase – Search by Sequence):
- Look for homologs in other priority pathogens
- Identify hits with very high sequence similarity (at least four priority pathogens)

↓

Protein Features of Homologs (in each pathogen):
- Essentiality
- Centrality
- Non-homology to human proteome
- Non-paralog status
- Role in virulence
- Association with antimicrobial resistance (AMR)

↓

Broad-spectrum drug target candidates (e.g., LpxA) supported by combined evidence



Use Case 3:

Extend from Use Case 2 candidate (LpxA)

↓

Structure Search in ARKbase:
- Input: Two compounds reported as potential LpxA inhibitors

↓

Search Results:
- No hits obtained with known antibiotics in ARKbase
- Indicates absence of overlap with existing scaffolds

↓

Potential novel scaffolds for LpxA inhibition → supports drug discovery against multi-drug-resistant Gram-negative bacteria

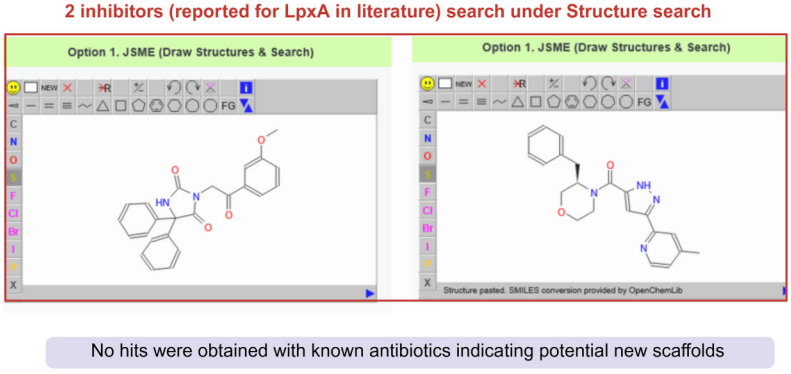


Figure 2. Schematic illustration of use cases highlighting the utilities of ARKbase.

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select a specific pathogen from the BPPL list, provide a valid Genome Accession Number (e.g. from NCBI), and include the corresponding accession URL. This ensures that all submitted genomic data are linked to publicly accessible and verifiable sources. For phenotype data, contributors must provide AST information for one or more antibiotics, including either MIC or disk diffusion (DD) values. These criteria help maintain the quality and consistency of community-submitted data. Once the data is evaluated for suitability for integration, all features will then be computed and integrated in ARKbase.

Future versions of ARKbase will include additional functionalities, namely, GWAS, MLST, transcriptome data analysis modules, and extensions to non-human isolates. The release cycle is planned on an annual basis.

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Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

ARKbase data, as described in the three core modules—Database module, Insights module, and Comparative Analysis module—are freely available at http://datascience.imtech.res.in/anshu/arkbase/. All data are provided on the ARKbase platform and also on https://doi.org/10.6084/m9.figshare.30329842.v2, and the source code used for annotations is provided at https://github.com/AB-DataScienceLab/ARKbase.

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